

COURSE TITLE : DESIGN OF JIGS & FIXTURES
COURSE CODE : 6304
COURSE CATEGORY : A
PERIODS / WEEK : 5
PERIODS / SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Introduction	19
2	Locations, Clamping ,Drill Bushes	18
3	Jigs	19
4	Fixtures	19
TOTAL		75

COURSE OUTCOME:

1. Understand the importance of jigs and fixtures
2. Understand the different degrees of freedom, methods of arresting and locations
3. Knowledge about clamping devices
4. Understanding the classification of drill bushes
5. Analyze the function of every element of different jigs used
6. Analyze the function of every element of different fixtures used

MODULE I

1.1.0 Understand the importance of jigs and fixtures

- 1.1.1 Definition of jigs, fixtures and its need
- 1.1.2 Describe Interchangeability
- 1.1.3 Understand the Difference between jigs and fixtures
- 1.1.4 Identify the Elements of jigs and fixtures
- 1.1.5 Enlist the Materials
- 1.1.6 Describe Advantages and disadvantages
- 1.1.7 Applications

MODULE II

2.1.0 Understand the different degrees of freedom, methods of arresting and locations

- 2.1.1 Definition of locations and its need
- 2.1.2 Explain Degrees of freedom, 3-2-1 principle
- 2.1.3 Types of locators
- 2.1.4 Describe redundant location
- 2.1.5 Explain Fool proofing

2.2.0 Knowledge about clamping devices

- 2.2.1 Definition of clamping and need
- 2.2.2 Requirements of good clamping system
- 2.2.4 Types of clamping devices

2.3.0 Understanding the classification of drill bushes

2.3.1 Introduction

2.3.2 Classification of drill bushes

MODULE III

3.1.0 Analyze the function of every element of different jigs used

3.1.1 Introduction

3.1.2 Describe the Constructional features

3.1.3 Types of jigs

3.1.4 Design considerations

3.1.5 Jig design for some simple components

MODULE IV

4.1.0 Analyze the function of every element of different fixtures used

4.1.1 Introduction

4.1.2 Describe Constructional features

4.1.3 Explain setting blocks and tenons

4.1.4 Types of fixtures

4.1.5 Design considerations

4.1.6 Fixture design for some simple components

CONTENT DETAILS

MODULE I

Definition of jigs- fixtures and its need-Interchangeability- Difference between jigs and fixtures- Elements of jigs and fixtures - Materials- Advantages and disadvantages-Applications

MODULE II

Principles of location – location point – types of locators – pins and studs – V block – cup and cone location points – adjustable locating points – special adjustable stops – location from finished holes in the work – Diamond pin locator – Cam operated 'V' locator – Quick action 'V' locator - Six point location of a three legged object – Location of a cylinder on a v-block- Degrees of freedom, 3-2-1 principle-

Principles of clamping – types of clamping – lever clamp – hinged clamp – two-way clamp – swinging clamp – wedge clamp – eccentric clamping arrangement – quick action clamp – Cam operated clamp – quarter turn screw – Toggle clamp – Pneumatic and hydraulic clamps – Washers - 'C' washer – spherical and flat washers. Requirements of good clamping system-

Introduction of drill bushes - Classification of drill bushes- Headless wearing press-fit bushing, Head wearing press-fit bushing, Headless liner bushing with renewable bushing, Head liner bushing with renewable bushings

MODULE III

Introduction-Describe the Constructional Features-Types of Jigs-Design Considerations-Jig design for some simple components

MODULE IV

Introduction-Describe Constructional features-setting blocks and tenons-Types of Fixtures-Design Considerations-Fixture design for some simple components

REFERENCE BOOKS

- 1 JIGS AND FIXTURES by P. H.Joshi
- 2 An introduction to Jig and Tool Design by M.H.A. Kempster